

PYTHON-TEST-2

MCQ

(½ X32)=16M

Q1. What is the primary purpose of the if statement in Python?

- a) To execute a block of code based on a condition b) To perform mathematical operations
c) To repeat a block of code d) To define a function

Q2. What will be the output of the following code snippet?

```
x = 10
```

```
if x > 5:
```

```
    print("x is greater than 5")
```

- a) x is greater than 5 b) x is less than 5 c) x is equal to 5 d) No output

Q3. Which keyword is used to execute a block of code if the condition in the if statement is false?

- a) else b) elif c) while d) for

Q4. What is the purpose of the elif statement in Python?

- a) To execute a block of code if the previous conditions are false b) To define a loop
c) To perform arithmetic operations d) To exit from a loop

Q5. What will be the output of the following code snippet?

```
x = 5
```

```
if x < 3:
```

```
    print("x is less than 3")
```

```
elif x == 3:
```

```
    print("x is equal to 3")
```

```
else:
```

```
    print("x is greater than 3")
```

- a) x is less than 3 b) x is equal to 3 c) x is greater than 3 d) No output

Q6. How can you execute multiple statements under a single if block in Python?

- a) Separate statements with a semicolon b) Indent the statements to the same level
c) Use the and keyword between statements d) Use the elif keyword

Q7. What will the following code snippet print?

```
x = 10
```

```
if x < 5:
```

```
    print("x is less than 5")
```

```
elif x > 15:
```

```
    print("x is greater than 15")
```

```
else:
```

```
    print("x is between 5 and 15")
```

- a) x is less than 5 b) x is greater than 15 c) x is between 5 and 15 d) No output

Q8. What is the purpose of the while loop in Python?

- a) To execute a block of code repeatedly until a condition is false
b) To execute a block of code a fixed number of times
c) To define a function d) To iterate over items in a sequence

Q9. What is the syntax for a while loop in Python?

- a) while condition: b) while condition(): c) while (condition): d) while loop condition:

Q10. How can you exit a loop prematurely in Python?

- a) Using the break statement b) Using the continue statement
c) Using the pass statement d) Using the exit function

Q11. What is the purpose of the for loop in Python?

- a) To execute a block of code repeatedly until a condition is false
b) To iterate over items in a sequence c) To execute a block of code a fixed number of times
d) To define a function

Q12. What is the syntax for a for loop in Python?

- a) for item in sequence:
b) for item in range(n):
c) for index in range(len(sequence)):
d) All of the above

Q13. How can you skip the current iteration of a loop and continue with the next iteration?

- a) Using the skip statement b) Using the pass statement
- c) Using the break statement d) Using the continue statement

Q14. What is the purpose of the range() function in Python?

- a) To generate a sequence of numbers b) To iterate over items in a sequence
- c) To define a function d) To execute a block of code repeatedly until a condition is false

Q15. What will the following code snippet print?

```
for i in range(3):
```

```
    print(i)
```

- a) 0 1 2 b) 1 2 3 c) 2 1 0 d) 3 2 1 0

Q16. What will be the output of the following code?

```
for i in range(1, 6):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```

- a) 1 2 b) 1 2 3 c) 1 2 4 5 d) 1 2 4

Q17. What will be the output of the following code?

```
for i in range(3):
```

```
    for j in range(3):
```

```
        print(i + j, end=' ')
```

```
    print()
```

- a) 0 1 2

```
1 2 3
```

```
2 3 4
```

- b) 0 1 2

```
1 2 3
```

```
2 3 4
```

3 4 5

c) 0 1 2

1 2 3

2 3 4

4 5 6

d) 0 1 2

2 3 4

4 5 6

Q18. What will be printed by the following code?

```
num = 5
```

```
while num > 0:
```

```
    print(num)
```

```
    num -= 1
```

```
    if num == 3:
```

```
        break
```

```
else:
```

```
    print("Done")
```

a) 5 4 3 2 1

b) 5 4

c) Done

d) 5 4 3

Q19. What is the output of the following code?

```
x = 10
```

```
if x > 5:
```

```
    print("A")
```

```
elif x > 7:
```

```
    print("B")
```

```
else:
```

```
    print("C")
```

- a) A b) B c) C d) A and B

Q20. What will be the output of the following code?

```
x = 5
```

```
while x > 0:
```

```
    print(x, end=" ")
```

```
    x -= 2
```

```
    if x == 1:
```

```
        break
```

```
else:
```

```
    print("Done")
```

- a) 5 3 1 b) 5 3 c) 5 3 Done d) 5 3 1 Done

Q21. What is the output of the following code?

```
for i in range(5):
```

```
    if i == 2:
```

```
        continue
```

```
    print(i, end=" ")
```

- a) 0 1 3 4 b) 0 1 2 3 4 c) 0 1 3 4 5 d) 0 1 2 3 4 5

Q22. What is the output of the following code?

```
num = 0
```

```
while num < 5:
```

```
    print(num)
```

```
    num += 1
```

```
else:
```

```
    print("Loop completed.")
```

- a) 0 1 2 3 4 b) 0 1 2 3 4 Loop completed. c) Loop completed. d) This code will result in an error.

Q23. What will be the output of the following code?

```
x = 10
```

```
if x > 5:
```

```
    print("Greater than 5")
if x > 8:
    print("Greater than 8")
if x > 12:
    print("Greater than 12")
else:
    print("Equal or less than 12")
```

a) Greater than 5

Greater than 8

Equal or less than 12

b) Greater than 5

Greater than 8

Greater than 12

Equal or less than 12

c) Greater than 5

Greater than 8

d) Equal or less than 12

Q24. What will be the output of the following code?

```
for i in range(3):
    for j in range(3):
        print(i * j, end=' ')
    print()
```

a) 0 0 0

0 1 2

0 2 4

b) 0 1 2

0 2 4

0 3 6

c) 0 0 0

0 1 2

0 2 4

0 3 6

d) 0 0 0

0 1 2

0 1 2

Q25. What will be the output of the following code?

```
num = 10
```

```
while num > 0:
```

```
    print(num)
```

```
    num -= 3
```

```
else:
```

```
    print("Loop completed.")
```

a) 10 7 4 1 Loop completed. b) 10 7 4 1 c) Loop completed. d) This code will result in an error.

Q26. What will the following code snippet print?

```
for i in range(3):
```

```
    pass
```

```
print(i)
```

a) 0 1 2 b) 1 2 3 c) 2 1 0 d) 3 2 1 0

Q27. What is the purpose of the pass statement in Python?

a) To exit from a loop prematurely b) To skip the current iteration of a loop

c) To execute a block of code if a condition is false d) To do nothing and act as a placeholder

Q28. What is the purpose of the else block in a loop in Python?

a) To execute if the loop encounters an error

b) To execute if the loop completes without encountering a break statement

c) To execute if the loop encounters a continue statement

d) To execute if the loop encounters a pass statement

Q29. What is the output of the following code?

```
x = 10
```

if x > 5:

 print("Hello")

elif x > 8:

 print("Hi")

else:

 print("Hey")

a) Hey b) Hi c) Hello d) Hello, Hi

Q30. What is the output of the following code?

for i in range(1, 6):

 if i % 2 == 0:

 print(i)

 continue

 print("*")

a) \n2\n\n4\n* b) \n2\n\n4\n*\n c) \n2\n\n*\n4\n d) \n\n2\n*\n4\n

Q31. What will be the output of the following code?

x = 5

while x > 0:

 print(x)

 x -= 1

else:

 print("Done")

a) 5\n4\n3\n2\n1\nDone b) Done\n5\n4\n3\n2\n1 c) 5\n4\n3\n2\n1 d) Done

Q32. What will be the output of the following code?

for i in range(1, 6):

 if i == 3:

 break

 print(i)

else:

 print("Loop completed.")

- a) 1\n2 b) 1\n2\n3\n4\n5\nLoop completed. c) 1\n2\n3\n4\n5 d) 1\n2\n3

2 MARKS

(2X5)=10M

1. what is python and Difference between .py file and .pyc file ?
- 2.what is dynamic type with example?
 - i.what is PEP8?
 - ii.what is interpreter ?
 - iii.how do you comment in python?
- 3.what are the datatypes in python?
 - i.what is python operators and types of operators?
- 4.what is flow control statements in python ?Difference b\w for and while loop?
- 5.Difference between python 2 and python 3?

4MARKS

1. Write the program to print right angular pattern.

20MARKS

1. Write the proqram ?

1.
E D C B A
D C B A
C B A
B A
A

2.
A 1
B 2 C 3
D 4 E 5 F 6
G 7 H 8 I 9 J 10

3.
G 1 E 4 C 9 A
F 1 D 4 B 9
E 1 C 4 A
D 1 B 4
C 1 A
B 1
A

4.
I 3 G 4 E 5 C
H 3 F 4 D C
G 3 E 4 C
E 3 D 4
E 3 C
D 3
C

5.
A
1 3
C E G
5 7 9 11
I K M O Q

1.)
EDCBA
DCBA
CBA
BA
A

```
1. for i in range(5, 0, -1):  
    for j in range(i, 0, -1):  
        print(chr(64 + j), end="")  
    print()
```

Row	(i)	j	Loop Values	Characters Printed
------------	------------	----------	--------------------	---------------------------

5		5, 4, 3, 2, 1	EDCBA
4		4, 3, 2, 1	DCBA
3		3, 2, 1	CBA
2		2, 1	BA
1		1	A

2. A1
B2C3
D4E5F6
G7H8I9J10

Ans:

char = 65 # ASCII value of 'A'

```

num = 1
for i in range(1, 5):
    for j in range(i):
        print(f"{chr(char)}{num}", end="")
        char += 1
        num += 1
    print()

```

3.
 G1E4C9A
 F1D4B9
 E1C4A
 D1B4
 C1A
 B1
 A

Ans:

```

letters = ['G', 'E', 'C', 'A']
nums = [1, 4, 9]
for i in range(7, 0, -1):
    for j in range(0, i, 2):
        if j // 2 < len(letters):
            print(letters[j // 2], end="")
        if j // 2 < len(nums):
            print(nums[j // 2], end="")
    print()

```

4.
 I3G4E5C
 H3F4DC
 G3E4C
 E3D4
 E3C
 D3
 C

Ans:

```
chars = ["I", "H", "G", "E", "D", "C"]
nums = [3, 4, 5]
for i in range(len(chars)):
    for j in range(0, len(chars) - i, 2):
        print(chars[j], end="")
        if j // 2 < len(nums):
            print(nums[j // 2], end="")
    print()
```

5.

A

13

CEG

57 9 11

IKMOQ

Ans:

```
print("A")
```

```
print("13")
```

```
print("CEG")
```

```
print("57 9 11")
```

```
print("IKMOQ")
```

4mark

EDCBA

DCBA

CBA

BA

A

```
for i in range(5, 0, -1):
```

```
    for j in range(i, 0, -1):
```

```
        print(chr(64 + j), end="")
```

```
    print()
```

functions in python

12-09-2024

1.create a function called add(),sub(),mul(),div()

-And get the input for a and b inside every function print the result.

2.Get a integer number from user and pass it to the function called findevenorodd().

Let the function print whether the number is even or odd.

3.Get a integer number from user and pass it to the function called findpassorfail().Let the function print whether pass or fail.(<35---fail)

4.Get input for a and b and pass it to the function called printrange().Let the function print numbers from a to b